Cohen writes that a phase-space formulation is possible such that provided

$$\langle \mathbf{H} \rangle = \int \psi^*(q)\mathbf{H}(q,p)\psi(q) = \iint h(q,p)\mathbf{F}(q,p)\ dq\ dp \tag{1}$$

where $\mathbf{F}(q,p)$ is the distribution. Cohen generalizes this distribution to be in the form

$$\mathbf{F}(q,p,t;f) = \frac{1}{4\pi^2}\iiint e^{-i\theta q - i\tau p + i\theta u} f(\theta,\tau,t)\psi^*\left(u - \frac{1}{2}\tau\hbar, t\right)\psi\left(u + \frac{1}{2}\tau\hbar, t\right)\ d\theta\ d\tau\ du \tag{2}$$

where $f(\theta,\tau,t)$ satisfies the relation such that

$$f(0,\tau,t) = f(\theta,0,t) = 1 \tag{3}$$

.

In this juncture, we choose a function f such that

$$f(\theta,\tau,t) = \sum_{n=0}^{\infty} a_n\left(\frac{1}{2}\theta\tau\hbar\right)^n = 1 + a_1\left(\frac{1}{2}\theta\tau\hbar\right) + a_2\left(\frac{1}{2}\theta\tau\hbar\right)^2 + a_3\left(\frac{1}{2}\theta\tau\hbar\right)^3 + \cdots \tag{4}$$

for which we solve for Eq. 2. As per the condition, we set $a_0 = 1$ so that when $\theta, \tau = 0$, then $f(\theta,\tau,t) = 1$. The constants are to be determined later for $n = 1, 2, 3, \ldots$. We find

$$\begin{aligned}\mathbf{F}(q,p,t;f) = \frac{1}{4\pi^2}\iiint e^{-i\theta q - i\tau p + i\theta u}&\left(1 + a_1\left(\frac{1}{2}\theta\tau\hbar\right) + a_2\left(\frac{1}{2}\theta\tau\hbar\right)^2 + \cdots\right)\\ &\times \psi^*\left(u - \frac{1}{2}\tau\hbar, t\right)\psi\left(u + \frac{1}{2}\tau\hbar, t\right)\ d\theta\ d\tau\ du\end{aligned} \tag{5}$$

and rearrange for all functions to group the θ's to one integral.

$$\begin{aligned}\mathbf{F}(q,p,t;f) = \frac{1}{4\pi^2}\iint e^{-i\tau p}&\psi^*\left(u - \frac{1}{2}\tau\hbar, t\right)\psi\left(u + \frac{1}{2}\tau\hbar, t\right)\\ &\times\left[\int e^{i\theta(u-q)}\left(1 + a_1\left(\frac{1}{2}\theta\tau\hbar\right) + a_2\left(\frac{1}{2}\theta\tau\hbar\right)^2 + \cdots\right)\ d\theta\right]\ d\tau\ du\end{aligned} \tag{6}$$

We compute for the continuous integrals. We allow

$$\int e^{i\theta(u-q)}\ d\theta = 2\pi\delta(u-q) \tag{7}$$

$$a_1\frac{1}{2}\tau\hbar\int \theta e^{i\theta(u-q)}\ d\theta = -2\pi i a_1\left(\frac{1}{2}\tau\hbar\right)\delta'(u-q) \tag{8}$$

$$a_2\left(\frac{1}{2}\tau\hbar\right)^2\int \theta^2 e^{i\theta(u-q)}\ d\theta = -2\pi a_2\left(\frac{1}{2}\tau\hbar\right)^2\delta''(u-q) \tag{9}$$

$$\vdots \tag{10}$$

$$a_n\left(\frac{1}{2}\tau\hbar\right)^n\int \theta^n\, e^{i\theta(u-q)}\, d\theta = 2\pi a_n\left(\frac{1}{2}\tau\hbar\right)^n\frac{1}{i^n}\,\delta^{(n)}(u-q) \tag{11}$$

and thus

$$\begin{aligned}\mathbf{F}(q,p,t;f) = \frac{1}{2\pi}\iint e^{-i\tau p}&\psi^*\left(u - \frac{1}{2}\tau\hbar, t\right)\psi\left(u + \frac{1}{2}\tau\hbar, t\right)\\ &\times\left[\delta(u-q) - a_1 i\left(\frac{1}{2}\tau\hbar\right)\delta'(u-q) - a_2\left(\frac{1}{2}\tau\hbar\right)^2\delta''(u-q) + \cdots\right]\ d\tau\ du\end{aligned} \tag{12}$$

The bracketed term could be simplied to an exponential expansion such that

$$e^{-\frac{1}{2}\tau\hbar\frac{d}{du}}\delta(u-q) = \delta\left(u - q - \frac{1}{2}\tau\hbar\right) \tag{13}$$

if the coefficients $a_0, a_1, a_2, a_3, \ldots$ are picked to be $a_n = (-1)^n/n!$ specifically. The equation thus allows such that

$$\mathbf{F}(q,p,t;f) = \frac{1}{2\pi} \iint e^{-i\tau p} \psi^* \left(u - \frac{1}{2}\tau\hbar, t\right) \psi \left(u + \frac{1}{2}\tau\hbar, t\right) \delta \left(u - q - \frac{1}{2}\tau\hbar\right) \, d\tau \, du \tag{14}$$

Recalling that

$$\int f(x)\delta(x-a) \, dx = f(a) \tag{15}$$

and thus

$$\boxed{\mathbf{F}(q,p,t;f) = \frac{1}{2\pi} \int e^{-i\tau p} \psi^* (q,t) \, \psi \left(q + \tau\hbar, t\right) \, d\tau} \tag{16}$$

arriving at the **general distribution** for quantization.